# Problem-Based Learning (PBL)

AIDS Department — Evaluation Rubrics

Students must independently choose a real-world problem statement and a relevant dataset from Kaggle or another reliable source. The project should demonstrate the applicable data science and Machine Learning lifecycle stages covered in the curriculum.

| Sr. No. | Criteria | Marks | What to Evaluate |
|---|---|---|---|
| 1 | Problem Definition & Business Understanding | 10 | Clear problem statement, objectives, stakeholders, business relevance, and KPIs |
| 2 | Dataset & Data Understanding | 10 | Dataset relevance, feature understanding, data dictionary, data types, and data quality |
| 3 | Data Preprocessing & Cleaning | 10 | Handling missing values, duplicates, outliers, encoding, scaling, and appropriate transformations |
| 4 | Exploratory Data Analysis (EDA) | 10 | Statistical analysis, meaningful visualizations, patterns, correlations, and insights |
| 5 | Feature Engineering / Selection | 10 | Creation or selection of relevant features, handling data leakage, and justification |
| 6 | Model Selection & Implementation | 15 | Appropriate AI/ML algorithm, implementation, train-test split, pipeline, and methodology |
| 7 | Model Evaluation & Validation | 15 | Appropriate evaluation metrics, validation strategy, confusion matrix or other relevant metrics, and interpretation |
| 8 | Business Insights & Recommendations | 10 | Conversion of data analysis and model results into actionable recommendations |
| 9 | Documentation & Presentation | 10 | Report quality, PPT, code organization, GitHub/README, project demonstration, presentation, and Q&A |
| | **Total** | **100** | |

## Additional Expectations for AIDS Students

- Choose an original real-world problem statement and a suitable dataset from Kaggle or another reliable source.
- Demonstrate the applicable data science and Machine Learning lifecycle stages covered in the curriculum.
- Justify the choice of algorithms, preprocessing techniques, features, and evaluation metrics.
- Use statistical analysis and visualizations to support conclusions drawn from the data.
- Explain the practical relevance of findings and each team member’s individual contribution.

**Note: A working deployment may be demonstrated as part of the project presentation where applicable, although it is not a separate marks category in this rubric.**